

Satellite-Based Monitoring of Methane Emissions from China's Rice Hub

Ruosi Liang, Yuzhong Zhang,* Qiwen Hu, Tingting Li, Shihua Li, Wenping Yuan, Jialu Xu, Yujia Zhao, Peixuan Zhang, Wei Chen, Minghao Zhuang, Guofeng Shen, and Zichong Chen



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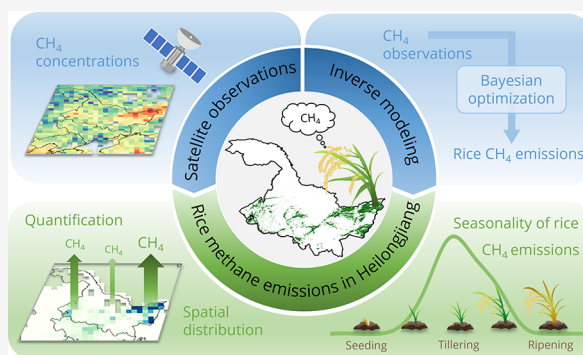
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ABSTRACT: Rice cultivation is one of the major anthropogenic methane sources in China and globally. However, accurately quantifying regional rice methane emissions is often challenging due to highly heterogeneous emission fluxes and limited measurement data. This study attempts to address this issue by quantifying regional methane emissions from rice cultivation with a high-resolution inversion of satellite methane observations from the Tropospheric Monitoring Instrument (TROPOMI). We apply the method to the largest rice-producing province (Heilongjiang) in China for 2021. Our satellite-based estimation finds a rice methane emission of 0.85 (0.69–1.03) Tg a^{-1} from the province or an average emission factor of 22.0 (17.8–26.6) $\text{g m}^{-2} \text{a}^{-1}$ when normalized by rice paddy areas. The satellite-based analysis reveals a 2 to 4 times lower bias in widely used global and national inventories, which lack up-to-date regional information. The inversion reduces the uncertainty of regional rice emissions by 73% relative to bottom-up estimates based on field flux measurements. The satellite inversion also shows that the highest rice methane emissions occur in June during the tillering stage of rice, decreasing toward ripening, indicating that the predominant water management practice in the region involves drainage and intermittent flooding after initial flooding. Process-based modeling further suggests that this practice can lead to a reduction of methane emissions by more than 50% compared to continuous flooding of rice paddies and natural wetlands.

KEYWORDS: methane emissions, rice, inverse modeling, satellite, Northeast China



1. INTRODUCTION

Rice cultivation is one of the most important sources of atmospheric methane, accounting for ~8% of global anthropogenic methane emissions.¹ Its contribution is anticipated to further increase in coming decades driven by the expansion of rice paddies coupled with climate warming.^{2–5} Notably, in China, the largest rice producer globally, rice cultivation alone contributes to ~20% of anthropogenic methane emissions. In November 2023, China published the Methane Emission Control Action Plan, calling for active control of methane emissions before 2030.⁶ This plan delineates various measures, such as improved water and fertilizer management practices and the selection of low-emission, high-yield cultivars, to reduce emissions from rice cultivation.

Methane is generated in rice paddies through the interplay between methanogens, which produce methane, and methanotrophs, which consume methane.⁷ The emission flux is influenced by both environmental factors (e.g., soil temperature, pH, and carbon content) and agricultural practices (e.g., water and fertilizer management),^{8–10} resulting in substantial spatial and temporal variability. Generally, flooding in rice

paddies leads to the decomposition of soil organic matter and methane production, and ~90% of that is emitted from soil to the atmosphere through the aerenchyma tissue of rice plants, and the rest are through diffusion or ebullition.^{11,12} Thus, the seasonal variability of rice methane fluxes is controlled by rice growth and development and is largely affected by water and fertilizer management.^{13,14} Four typical temporal patterns for the growing season emission have been summarized in a recent meta-analysis based on field flux measurements worldwide.¹⁵

Accurately quantifying rice methane emissions and assessing emission reduction measures at the regional scale is often challenging.¹⁶ Bottom-up estimates typically rely on emission factor data obtained from field flux measurements or process-based modeling.¹⁷ However, the heterogeneity of rice emissions limits the representativeness of often sparse rice

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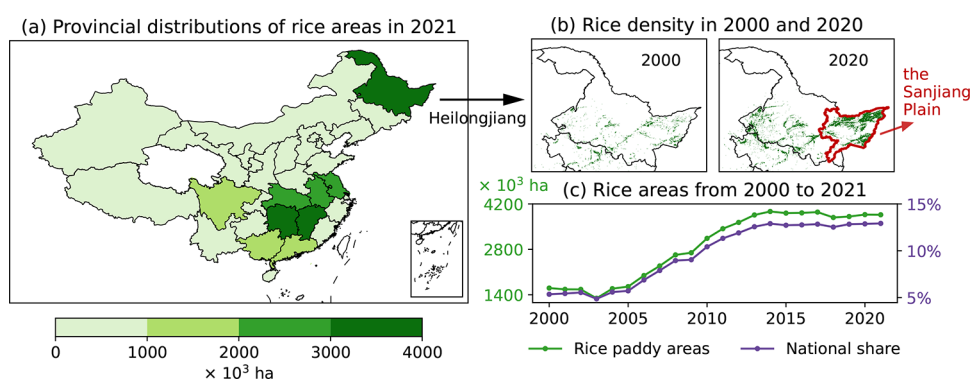


Figure 1. Rice cultivation areas during 2000–2021 in China and Heilongjiang province. (a) Provincial rice paddy areas in 2021.⁴² White shading represents no available data. (b) Rice paddy density in Heilongjiang in 2000 and 2020.⁴³ The extent of the Sanjiang Plain, located in the eastern part of Heilongjiang, is enclosed by a red line. (c) Time series of rice cultivation areas in Heilongjiang and its share of the national total between 2000 and 2021.

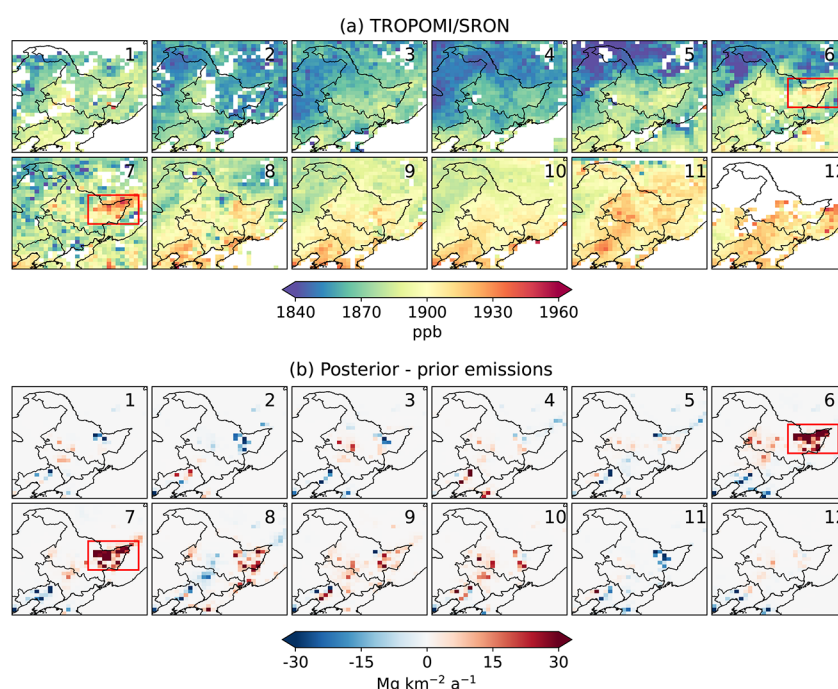


Figure 2. Satellite observations and inversion results from the base inversion (SRON-E1). (a) Monthly average TROPOMI/SRON methane columns and (b) monthly emission corrections.

field sites, leading to large uncertainties at the regional scale.^{17–19} Process-based models are also subject to uncertain parameters and often depend on field flux measurements for calibration.^{12,20–23} Existing bottom-up estimates of China's total rice methane emissions range from 2.3 to 30.0 Tg a^{-1} with great regional discrepancies.^{24,25}

Satellite observations offer an alternative approach to constrain regional-scale rice methane emissions. Previous studies have reported satellite-based estimates of rice methane emissions at global or country levels at a coarse resolution.^{26–31} However, the constraints on rice emissions provided by these studies were usually weak owing to the limited coverage of high-quality satellite observations (e.g., SCIAMACHY and GOSAT). The advent of the TROPOMI sensor on the Sentinel 5 Precursor (SSP) satellite, launched in 2017, has enabled more fine-grained characterization of methane fluxes on smaller regional scales.^{32–34} TROPOMI observations have been employed to monitor regional methane emissions,

but the focus has been primarily on the energy and waste sectors.^{35–41}

Here, we evaluate monthly methane emissions from rice cultivation in Heilongjiang province using TROPOMI satellite observations. Located in Northeast China, Heilongjiang is the country's largest rice-producing province. As of 2021, the province produced 30,000 kt rice (14% of the national total, Figure S1) with an area of ~ 3800 kha rice fields (13% of the national total, Figure 1a), which is 2.4 times of its levels in 2000, owing to a rapid expansion of rice cultivation in the Sanjiang Plain in the 2010s⁴² (Figure 1b,c). Our analysis assesses rice methane emissions in this critical rice-producing region regarding their magnitude, spatial distribution, and seasonality. Furthermore, we compare our top-down results with bottom-up estimates, including inventories, process-based models, and field flux measurements.

2. DATA AND METHODS

2.1. Satellite Observations. We use column-averaged dry-air methane mole fractions (XCH_4) retrieved from the TROPOMI instrument onboard the sun-synchronous Sentinel 5 Precursor (S5P) satellite for the inversion.^{32–34,44} The TROPOMI measurements were taken at a resolution of 5.5 km $\times$ 7 km at nadir (since August 2019) with a $\sim$ 13:30 local overpass time. To quantify methane emissions and better characterize associated uncertainties, we use TROPOMI XCH_4 products from two retrieval algorithms: (1) The Netherlands Institute for Space Research (SRON) operational TROPOMI product v02.04.00⁴⁵ (hereafter referred to as TROPOMI/SRON), which is based on the SRON RemoTeC-S5P CH_4 scientific product v19_446,⁴⁴ and (2) the scientific TROPOMI/WFMD product v1.8 (hereafter referred to as TROPOMI/WFMD).⁴⁶ For TROPOMI/SRON, we use high-quality data with quality flag “qa_value = 1”. In addition, we apply a blended albedo filter with a threshold of 0.8 for ice/snow scenes from October to April^{47,48} (Figure 2a). We have shown that TROPOMI/SRON agrees well with GOSAT in Northeast China in a previous study.⁴⁹ For TROPOMI/WFMD, we use high-quality data with “xch4_quality_flag = 0” (Figure S2). TROPOMI/WFMD shows good agreement with TROPOMI/SRON and provides slightly better data coverage.⁵⁰ A comparison of the two data sets in our study domain shows a small systematic difference (TROPOMI/SRON–TROPOMI/WFMD = -3.0 ppb) and consistent spatial and temporal variations ($R = 0.84$) (Figure S3). The spatial distribution of daily gridded observation numbers is shown in Figure S4.

2.2. Atmospheric Transport Simulation and Prior Estimates. We use the nested version of GEOS-Chem v12.9.3 as the forward model for the inversion to characterize the relationship between surface emissions and column methane concentrations. The simulation is performed on a $0.5^\circ \times 0.625^\circ$ grid over Northeast China ($34^\circ N$ – $59^\circ N$, $114^\circ E$ – $142^\circ E$) with 47 vertical layers in the atmosphere and is driven by MERRA-2 meteorological fields from the NASA Global Modeling and Assimilation Office (GMAO).⁵¹ The initial concentration fields on January 1, 2021 and 3 hly boundary conditions for the simulation domain are generated from a global $4^\circ \times 5^\circ$ GEOS-Chem simulation. We enlarge the simulation domain by 5° in four directions beyond the study region ($39^\circ N$ – $54^\circ N$, $119^\circ E$ – $137^\circ E$) and aggregate these grid cells into 10 spatial clusters (Figure S5). To better account for the background error, the inversion optimizes emissions from these 10 clusters and concentration biases at 4 boundaries, together with methane emissions from the study region resolved by 792 native grid cells. Similar configurations have been applied in previous regional inverse studies.^{39,40}

The inverse analysis also requires prior information. We use EDGAR v6.0⁵² for anthropogenic methane emissions (E0) (Figure S6). We note that the main rice-producing region of Heilongjiang, the Sanjiang Plain, does not stand out in EDGAR v6.0 rice emissions. This is because EDGAR v6.0 uses a static global rice distribution for 2000⁵³ and therefore cannot capture the rapid increase of rice cultivation in the Sanjiang Plain since 2000 (Figure 1). To remedy this, we redistribute rice emissions in EDGAR v6.0 with updated spatial information on rice distributions⁴³ (E1) (Figure S7). For natural sources, we use wetland emissions from the ensemble average of WetCHARTs version 1.0,⁵⁴ biomass burning emissions from the Global Fire Emissions Database version 4

(GFED4),⁵⁵ termite emissions from Fung et al.,⁵⁶ and geological emissions from Maasakkers et al.⁵⁷

The estimation of rice emissions is additionally impacted by the uncertainty in the prior distribution of other significant emission sources. To assess this influence, we conduct sensitivity inversions by substituting the prior inventories for coal (E2) or wetlands (E3). The alternative coal inventory is taken from GFEI v2,⁵⁸ which incorporates the spatial information on coal mines in China provided by Sheng et al.⁵⁹ (Figure S8). The alternative wetland emission inventory is obtained from VISIT⁶⁰ (Figure S9).

2.3. Inverse Analysis. We perform inversions to estimate spatially resolved monthly methane emissions from Northeast China in 2021. Figure S10 illustrates the procedure of the inversion, which draws information from both satellite observations and prior estimates.^{61,62} Mathematically, the inversion minimizes a cost function $J(\mathbf{x})$, defined as

$$J(\mathbf{x}) = (\mathbf{x} - \mathbf{x}_A)^T \mathbf{S}_A^{-1} (\mathbf{x} - \mathbf{x}_A) + (\mathbf{y} - \mathbf{F}(\mathbf{x}))^T \mathbf{S}_O^{-1} (\mathbf{y} - \mathbf{F}(\mathbf{x})) \quad (1)$$

where $\mathbf{x}$ is the state vector to be optimized, consisting of (1) rice emissions, (2) nonrice emissions, and (3) boundary conditions on four sides of our domain, $\mathbf{x}_A$ is the prior estimates of $\mathbf{x}$, $\mathbf{y}$ is the observation vector (TROPOMI/SRON or TROPOMI/WFMD), and $\mathbf{F}(\mathbf{x})$ represents the forward model (GEOS-Chem). $\mathbf{S}_O$ and $\mathbf{S}_A$ are observational and prior error covariance matrices, respectively. Both $\mathbf{S}_O$ and $\mathbf{S}_A$ are taken to be diagonal. $\mathbf{S}_O$ is populated with the residual error method.⁶³ $\mathbf{S}_A$ is populated by assuming prior errors of 150% for monthly rice emissions, 75% for emissions of other sectors,⁴⁹ and 1% for boundary conditions.⁴⁹ The larger error for rice emissions is derived by assessing Northeast China field flux measurements reported in literature (Supporting Information 2), reflecting uncertainties in both the magnitude and the seasonality of rice emissions. We also use an alternative specification for $\mathbf{S}_A$ to test the sensitivity of posterior estimates to the error parameters, including perturbing the error standard deviations (SA-up and SA-down) and specifying a nondiagonal $\mathbf{S}_A$ with an error correlation length of 50 km (SA-nondiag) (Table S1).

Minimizing $J(\mathbf{x})$ at $\nabla_{\mathbf{x}} J(\mathbf{x}) = 0$ yields the optimal posterior solution:

$$\hat{\mathbf{x}} = \mathbf{x}_A + (\mathbf{K}^T \mathbf{S}_O^{-1} \mathbf{K} + \mathbf{S}_A^{-1})^{-1} \mathbf{K}^T \mathbf{S}_O^{-1} (\mathbf{y} - \mathbf{F}(\mathbf{x}_A)) \quad (2)$$

where $\mathbf{K} = \nabla_{\mathbf{x}} \mathbf{F}$ is the Jacobian matrix describing the sensitivity of observations $\mathbf{y}$ to the state vector $\mathbf{x}$. We perturb independently each element of $\mathbf{x}$ in a series of GEOS-Chem simulations to construct $\mathbf{K}$, column by column.

The averaging kernel matrix $\mathbf{A}$ that describes the sensitivity of the optimal solution to the true value:

$$\mathbf{A} = \frac{\partial \hat{\mathbf{x}}}{\partial \mathbf{x}} = \mathbf{I}_n - \hat{\mathbf{S}} \mathbf{S}_A^{-1}. \quad (3)$$

The trace of $\mathbf{A}$ is referred to as the degree of freedom for signals (DOFS), which represents the number of independent pieces of information constrained by an observing system.

We construct an ensemble of 11 inversions by perturbing observation data (TROPOMI/SRON and TROPOMI/WFMD), prior emissions (E0, E1, E2, and E3), and error statistics (Table S1). We use SRON-E1 (observation: TROPOMI/SRON; prior emission inventory: E1; default

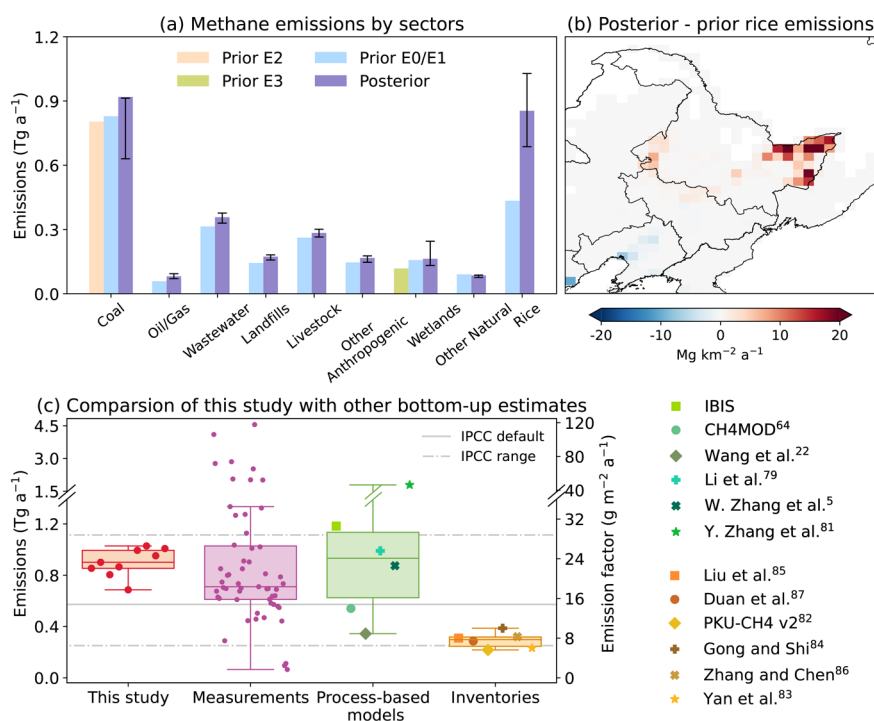


Figure 3. Rice methane emissions inferred from satellite in versions and comparison with varied bottom-up estimates. (a) Annual prior and posterior methane emissions from Heilongjiang attributed to source sectors. Error bars represent uncertainties of posterior estimates based on the inversion ensemble. (b) Spatial distribution of corrections to prior rice emissions during rice growth season (April–September). (c) Comparison of satellite-based inversions (red) with field flux measurements (purple), process models (green), national gridded inventories (yellow), and IPCC default values (gray lines). Supporting Information 2 summarizes information on these bottom-up studies.

S_A) as the base inversion and report the range of the ensemble (excluding SRON-E0 and SRON-E0-SA-up) as a measure of uncertainty in our estimate. We use SRON-E0 (the same as the baseline inversion except that E0 is used as the prior emission inventory) to quantify the biases induced by an incorrect prior spatial distribution of rice emissions.

2.4. Process Modeling by CH4MOD. We compare rice methane emissions inferred from satellite observations with those simulated by CH4MOD,⁹ a semiempirical process-based rice methane emission model, to understand the dominant water management practice in the study region and its impact on observed methane emissions.

The CH4MOD model simulates methane production in rice paddies and emissions to the atmosphere under various environmental conditions and agricultural practices.^{12,20} The rationale of the CH4MOD is that the methanogenic substrates are primarily derived from rice root substrates and added organic matter. The rates of methane production are determined by the availability of methanogenic substrates and the influence of environmental factors (including soil temperature, soil texture, and soil redox potential). Produced methane is emitted to the atmosphere via rice aerenchyma and bubbles. For a detailed description of the CH4MOD formulation, readers can refer to Huang et al.⁹ and Hu et al.⁶⁴

Inputs to the CH4MOD simulation include air temperature, soil sand fraction, harvest area, rice grain yield, water regimes, straw incorporation, crop rotation, and rice growth stage timing. The information on the input data sets (e.g., data source, spatial, and temporal resolution) is summarized in Table S2. Outputs of the CH4MOD simulation are daily methane emission fluxes on a $0.5^\circ \times 0.5^\circ$ grid. The CH4MOD model has been widely validated against measurements for the

period 1990–2019 from irrigated rice cultivations in various regions of the world, including China, the Philippines, Indonesia, Italy, and the United States, etc.^{5,9,64} The evaluation show that the model can achieve reasonable agreement with multiple field observations across China.^{5,9,23,64}

CH4MOD simulates the effect of water management on the soil redox potential (E_h), which subsequently affect methane production and emissions. The formulation of CH4MOD is designed to capture the rapid drop of E_h during the flooding course, the increase during the drainage course, and fluctuation during intermittent irrigation.⁹ Here, we consider six water management practices (i.e., F, F–D–F, F–D–F–M, M, F–D–M, and F–M, where F represents flooding, M intermittent irrigation, and D drainage). We then use simulation results under different water management practices to interpret the satellite-observed seasonality of rice emissions.

3. RESULTS AND DISCUSSION

3.1. Methane Enhancement over Intensive Rice Cultivation Areas. Figure 2a shows monthly maps of column-averaged methane mixing ratios over Northeast China from the TROPOMI/SRON product in 2021. Substantial seasonal enhancements of methane columns are found over the Sanjiang Plain. The enhancements are up to ~ 20 – 30 ppb relative to the background in June and July over the Sanjiang Plain (red rectangle in Figure 2a) and diminish toward wintertime. This seasonally enhanced methane mixing ratio over the Sanjiang Plain is also seen in the TROPOMI/WFMD product (Figure S2), confirming the robustness of the observations. But, this methane enhancement over the Sanjiang Plain cannot be captured by GOSAT satellite

observations because of sparse spatial coverage by GOSAT (Figure S11).

We compare the simulation based on the prior emission inventory E1 with TROPOMI/SRON XCH₄ to examine whether the observed methane enhancement can be reproduced by the simulation. The comparison shows that the prior simulation is lower than satellite observations by ~20 ppb in the Sanjiang Plain over June–July (Figure S12), indicating that the prior emission inventory is insufficient to capture variability in methane emissions, which we hope to correct through inverse analysis of satellite data.

The elevated methane concentration shown in Figure 2a corresponds to the Sanjiang Plain, a major rice-producing region in Heilongjiang province. The province ranks first in rice production and second in the cultivation area in China (14% of the country's rice production as of 2021) (Figures 1 and S1). The Sanjiang Plain, or more broadly Heilongjiang, is a relatively new hub of rice production in China, compared to South China. A rapid expansion of rice paddies occurred in Heilongjiang between 2005 and 2014, driven primarily by the Sanjiang Plain (Figure 1). Since the 1950s, natural wetlands in the Sanjiang Plain have been gradually reclaimed as drylands and then converted into rice paddies in the 2010s.^{65–67} This history explains that rice methane emissions are missing over this region in a few widely used gridded inventories such as EDGAR because these inventories depend on an outdated spatial distribution of rice paddies for the early 2000s.⁶⁸

3.2. Quantification of Rice Methane Emissions. Figure 2b shows the corrections of monthly prior total methane emissions on the $0.5^\circ \times 0.625^\circ$ grid inferred from the inversion of TROPOMI observations. Large upward corrections are made during the warm season (June–August) over the rice cultivation regions, mainly the Sanjiang Plain. This correction of methane emissions substantially reduces the model–observation mismatch over this region with mean biases decreased by 40% (Figure S12). Figure S13 characterizes the observation constraints of the inversion with the averaging kernel sensitivities (eq 3). The DOFS ranges from 1.0 in May to 3.3 in June over Heilongjiang Province, reflecting the seasonal variation in satellite observation density. The DOFS attributable to the rice sector is greater than 1 throughout the growing season (May–August), indicating a good observational constraint on regional-scale methane emissions from rice paddies.

Our inversion ensemble (Table S1) estimates a methane emission of 3.1 (2.8–3.3) Tg a^{−1} for Heilongjiang, compared to the prior estimate of 2.4 Tg a^{−1} (Figure 3a). About 60% of the upward correction is attributed to the rice sector, resulting in a rice methane emission of 0.85 (0.69–1.03) Tg a^{−1} for the province, with ~72% of that from the Sanjiang Plain. This estimation of rice methane emissions is twice as high as the prior estimate (0.43 Tg a^{−1}) by EDGAR v6.0 (Figure 3a,b) but closer to previous top-down estimates.^{49,69} Compared with the base inversion (SRON-E1: 0.85 Tg a^{−1}), sensitivity inversions show that our result is insensitive to the prior distribution of coal mine methane emissions (another major methane sector in Heilongjiang) (SRON-E2: 0.9 Tg a^{−1}) and wetland methane emissions (SRON-E1-wet: 0.87 Tg a^{−1}; SRON-E3: 0.8 Tg a^{−1}) and the choice of satellite product (WFMD-E1: 0.95 Tg a^{−1}; WFMD-E2: 1.01 Tg a^{−1}), and moderately sensitive to the specification of error statistics (SRON-E1-SA-down: 0.69 Tg a^{−1}; SRON-E1-SA-up: 0.99 Tg a^{−1}; SRON-SA-nondiag: 1.03 Tg a^{−1}) (Figure S14).

We show that employing the original EDGAR v6.0 inventory (E0; outdated rice spatial distribution) as the prior information (SRON-E0 and SRON-E0-SA-up) results in an underestimation of regional methane emissions attributable to the rice sector (SRON-E0: 0.56 Tg a^{−1}; SRON-E0-SA-up: 0.59 Tg a^{−1}), underscoring the importance of a correct prior spatial distribution in the regional methane inversion (Figure S14). EDGAR v6.0 spatially allocates national rice emissions based on a global distribution of rice paddies compiled for the year 2000,⁶⁸ thus significantly underrepresenting rice methane emissions in the Sanjiang Plain, which only became an important rice producer in the 2010s. A few other gridded inventories have similar issues (e.g., Regional Emission inventory in Asia v2.1 (REAS v2.1)⁷⁰).

3.3. Evaluation of Bottom-Up Estimations. We compile bottom-up rice methane studies for Northeast China, including field flux measurements,^{10,71–78} process models (three models: CH4MOD, IBIS, and DNDC),^{5,22,64,79–81} and global/national gridded emission inventories^{82–87} (Supporting Information 2) and compare our satellite-based results with these bottom-up studies. To facilitate comparison, we compute satellite-based region-average rice emission factors by normalizing regional methane emissions with the reported rice paddy area for 2021 (3876.39 kha).⁴²

Our inversions indicate an annual mean emission factor of 22.0 (17.8–26.6) g m^{−2} a^{−1}, which generally aligns with the central estimates of field flux measurements (mean: 26.5 g m^{−2} a^{−1}; median: 18.4 g m^{−2} a^{−1}) and process models (mean: 24.7 g m^{−2} a^{−1}; median: 24.1 g m^{−2} a^{−1}) and is within the range of the default emission factor (6.5–28.8 g m^{−2} a^{−1}) for East Asia given by the latest IPCC Guidelines⁸⁸ (Figure 3c).

However, both field flux measurements and process models show large variability in their estimates (Figure 3c). The emission factor based on field flux measurements ranges from 1.7 to 117.8 g m^{−2} a^{−1} (SD: ~22.8 g m^{−2} a^{−1}, ~86%) and that based on process models from 9.4 to 53.7 g m^{−2} a^{−1} (SD: ~13.2 g m^{−2} a^{−1}, ~53%). In comparison, our satellite-based analysis derives from our inversion ensemble an SD of 13% for the regional rice emission factor, indicating a 73% reduction in the variability relative to rice field flux measurements and a 40% reduction relative to process-based models (Figure 3c). The large variation in these bottom-up estimates reflects the challenge of representing rice emissions over a large region with sparse local measurements. Complex mechanisms and incomplete input data also add difficulty to process modeling. Our findings highlight that satellite observations serve as valuable supplements to field measurements and process modeling to reduce the uncertainty in quantifying regional rice emissions.

Moreover, our estimate for the rice methane emission factor is significantly higher than that employed in global and national gridded inventories for this region (i.e., EDGAR, PKU) (4.8–10.0 g m^{−2} a^{−1}). Small variability among these global and national inventories (4.8–10.0 g m^{−2} a^{−1}) is misleading because most of these studies are based on the same source for regional emission factors and thus are not independent. For example, EDGAR⁵³ and PKU-v2⁸² take emission factors from Yan et al.,⁸³ in which the emission factor for northern China is determined based on only one available data point. Gong and Shi,⁸⁴ Liu et al.,⁸⁵ and Zhang and Chen⁸⁶ use the emission factor from the same data set based on an early process model with no validation data in Northeast China.^{89–91} Our analysis identifies that these global and

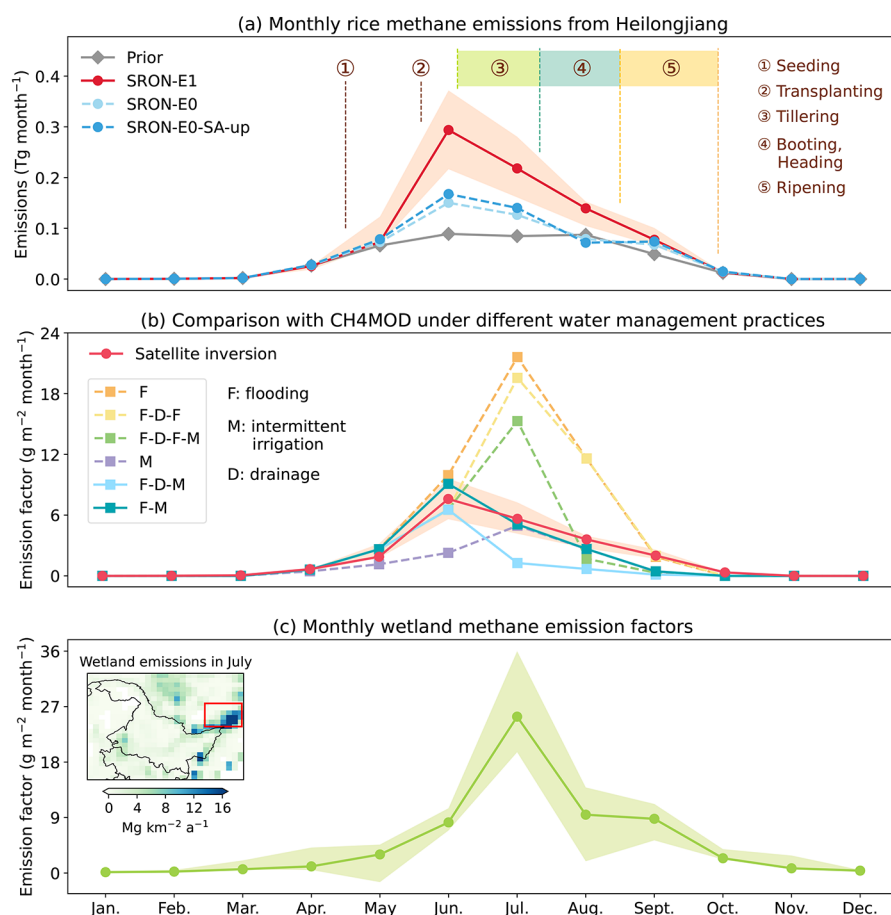


Figure 4. Monthly rice methane emissions from Heilongjiang province. (a) Prior and posterior (monthly rice methane emissions from Heilongjiang). Rice growth stages are denoted. Gray solid line represents prior estimates (EDGAR v6.0). Red solid line represents base posterior emissions (SRON-E1), and red shadings are the range of posterior estimates from sensitivity inversions (Table S1). Blue dashed lines represent inversion results based on an incorrect spatial distribution of rice methane emissions. (b) Comparison between the satellite inversion and CH4MOD simulations under six different water management scenarios. (c) Monthly wetland methane flux over a wetland-dominant region in Russia neighboring the Sanjiang Plain. Posterior methane emissions are normalized by wetland areas from GLOBCOVER data set (<http://due.esrin.esa.int>). The solid line shows the results of the base inversion and shadings represent the range of the inversion ensemble. The region is shown with a red rectangle in the inset map.

national inventories underestimate rice methane emissions in Northeast China because inadequate information is used for determining emission factors of this region.

3.4. Observed Emission Seasonality Linked to Water Management Practices. Figure 4a shows the monthly evolution of rice methane emissions inferred from the inversion throughout different growth stages. Methane emissions from rice cultivation start to increase in April and May during seeding and transplanting, peak in June during the early phase of tillering, and then gradually decrease to a low level toward harvest. This seasonality is distinct from the prior information given by EDGAR v6.0, which shows a sustained methane emission throughout the growing season (Figure 4a). In contrast, the inferred seasonality aligns with field flux measurements taken at an experiment site on the Sanjiang Plain (47°35'N, 133°30'E), although the magnitude of methane emissions at this site is a factor of 1.5–3 higher than the regional average derived by the inversion (Figure S15). Similar seasonality of rice methane emissions is also observed at other sites in Northeast China.^{71,73,75–77}

Previous studies have shown that both the magnitude and the seasonality of methane emissions from rice paddies are

closely linked with water management practices.¹⁵ Methane emission fluxes from rice paddies typically rise following seeding or transplanting and remain elevated throughout the growing season under conditions of continuous flooding. Implementing intermittent irrigation or draining the rice field for a brief period during the midseason generally reduces methane emissions.¹⁴ Such an impact of water management can be quantitatively simulated by process models. However, the information on the dominant practices is often difficult to obtain for process modeling. Comparing satellite-derived seasonality in rice emissions with those simulated by process models can provide information about the dominant water management practices over a large region.

Figure 4b shows the seasonal evolution of rice methane emissions simulated by the CH4MOD model under various water management practices. The seasonality inferred from satellite observations closely follows CH4MOD simulations under flooding–drainage–intermittent irrigation (F–D–M) or flooding–intermittent irrigation (F–M), with the maximum emissions in June, suggesting that F–D–M or F–M are likely the dominant water management practices in the region. Compared to continuous flooding (F), the practice with the

largest emissions, our analysis indicates that the dominant water management practices (F–D–M and F–M) in North-east China have reduced methane emissions by about 50% during the growing season and by 70% in July.

Interestingly, our inversion also finds substantial methane emissions from a wetland-dominant region neighboring the Sanjiang Plain (Figure 4c) during June–August with the maximum in July, similar to CH₄MOD simulations under continuous flooding (F) (Figure 4b). This natural wetland region in Russia is within the same river basin of the Sanjiang Plain and thus has similar environmental conditions. Our result suggests that natural wetlands in this region may resemble continuously flooded rice paddies in terms of methane emissions. Indeed, our inversion infers a high methane emission factor for wetland emissions from this region (60.5 (57.9–66.7) g m^{−2} a^{−1}), comparable to rice methane emission factors predicted by the CH₄MOD simulation under continuous flooding. The Sanjiang Plain has experienced extensive transformation from natural wetlands to drained croplands and rice paddies since the 1950s.^{72,79,92} This observation-derived information on emission factors for different land types is useful for accurately evaluating the overall impact of land use change on methane emissions.^{93,94}

3.5. Summary and Outlook. Mitigation of greenhouse gas emissions in the agricultural sector is a great challenge.⁹⁵ Balancing the impact on climate change with ensuring food security presents a substantial trade-off.⁹⁶ Moreover, the complex influence of multiple environmental and agricultural factors on greenhouse gas emissions makes it crucial to assess the effects of control measures in local contexts.⁹⁷ Accurate quantification of methane emissions and their response to mitigation measures, particularly at the regional level, is the foundation for informing effective strategies tailored to specific regions.

Traditionally, this quantification has been carried out through the bottom-up approach. However, the approach suffers from large uncertainties because of the intricate nature of rice emissions and the scarcity of high-quality, region-specific data required for calculations (e.g., environmental factors and agricultural practices).^{16,98} In addition, the bottom-up method often yields untimely results, as it involves time-consuming and costly data collection processes.⁹⁹

Our study showcases the potential of satellite-based top-down analysis as a powerful means to assess and enhance bottom-up results, supplying robust data to promptly inform climate and agricultural actions. Satellite-based quantification in this study leads to improved precision (reducing uncertainties in regional estimates relative to bottom-up methods based on field flux measurement or process models), accuracy (correcting biases in widely used emission inventories), and seasonality characterization (identifying the highest flux during the early phase of tillering) relative to bottom-up estimates. More importantly, we show that the satellite-based approach is sensitive to key parameters targeted by control measures, such as agricultural practices, thus demonstrating its ability to offer timely evaluation and feedback to stakeholders.

The inversion framework presented in our work can be extended to other rice cultivation regions in China and globally. We can further enhance the capability of this framework, by integrating satellite observations with ground-based stations and mobile measurements, forming a comprehensive monitoring network to support sustainable rice production.

■ ASSOCIATED CONTENT

Data Availability Statement

Results generated by this study are available at 10.57760/sciencedb.08588. The TROPOMI/SRON product is from <https://s5phub.copernicus.eu/dhus> (last access: 12 July 2023) (now available at <https://browser.dataspace.copernicus.eu/>). The TROPOMI/WFMD product is from https://www.iup.uni-bremen.de/carbon_ghg/products/tropomi_wfmd/ (last access: 26 August 2022). The spatial distribution of 2020 rice paddy areas is from 10.5281/zenodo.5555721.

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.est.4c09822>.

Details of the inversion ensemble; input data used in CH₄MOD; map of province-level rice productions; additional spatial information on satellite observations; details on inversion method (clusters, prior emissions by sectors, and inversion modeling framework); differences between TROPOMI observations and prior/posterior simulations; averaging kernel sensitivities and DOFS for the base inversion; all posterior emissions from the inversion ensemble; comparison with rice field flux measurements; and additional references (PDF)

Data set of IPCC default values, rice field flux measurements, inventories, and top-down estimates (XLSX)

■ AUTHOR INFORMATION

Corresponding Author

Yuzhong Zhang — Key Laboratory of Coastal Environment and Resources of Zhejiang Province, School of Engineering, Westlake University, Hangzhou, Zhejiang 310030, China; Institute of Advanced Technology, Westlake Institute for Advanced Study, Hangzhou, Zhejiang 310024, China; orcid.org/0000-0002-3548-8056; Email: zhangyuzhong@westlake.edu.cn

Authors

Ruosi Liang — College of Environmental and Resource Sciences, Zhejiang University, Hangzhou, Zhejiang 310058, China; Key Laboratory of Coastal Environment and Resources of Zhejiang Province, School of Engineering, Westlake University, Hangzhou, Zhejiang 310030, China; Institute of Advanced Technology, Westlake Institute for Advanced Study, Hangzhou, Zhejiang 310024, China; orcid.org/0000-0003-0479-5288

Qiwen Hu — School of Atmospheric Sciences, Guangdong Province Data Center of Terrestrial and Marine Ecosystems Carbon Cycle, Sun Yat-sen University, Zhuhai, Guangdong 510245, China

Tingting Li — State Key Laboratory of Atmospheric Boundary Layer Physics and Atmospheric Chemistry, Institute of Atmospheric Physics, Chinese Academy of Sciences, Beijing 100029, China; orcid.org/0000-0002-6338-2769

Shihua Li — School of Atmospheric Sciences, Guangdong Province Data Center of Terrestrial and Marine Ecosystems Carbon Cycle, Sun Yat-sen University, Zhuhai, Guangdong 510245, China

Wenping Yuan — College of Urban and Environmental Sciences and Institute of Carbon Neutrality, Peking University, Beijing 10871, China

Jialu Xu – Joint International Research Laboratory of Catastrophe Simulation and Systemic Risk Governance and School of National Safety and Emergency Management, Beijing Normal University, Zhuhai 519087, China

Yujia Zhao – College of Environmental and Resource Sciences, Zhejiang University, Hangzhou, Zhejiang 310058, China; Key Laboratory of Coastal Environment and Resources of Zhejiang Province, School of Engineering, Westlake University, Hangzhou, Zhejiang 310030, China; Institute of Advanced Technology, Westlake Institute for Advanced Study, Hangzhou, Zhejiang 310024, China

Peixuan Zhang – Key Laboratory of Coastal Environment and Resources of Zhejiang Province, School of Engineering, Westlake University, Hangzhou, Zhejiang 310030, China; Institute of Advanced Technology, Westlake Institute for Advanced Study, Hangzhou, Zhejiang 310024, China; Fudan University, Shanghai 200433, China

Wei Chen – College of Environmental and Resource Sciences, Zhejiang University, Hangzhou, Zhejiang 310058, China; Key Laboratory of Coastal Environment and Resources of Zhejiang Province, School of Engineering, Westlake University, Hangzhou, Zhejiang 310030, China; Institute of Advanced Technology, Westlake Institute for Advanced Study, Hangzhou, Zhejiang 310024, China

Minghao Zhuang – State Key Laboratory of Nutrient Use and Management, College of Resources and Environmental Sciences, Key Laboratory of Plant-Soil Interactions, Ministry of Education, China Agricultural University, Beijing 100193, China

Guofeng Shen – College of Urban and Environmental Sciences and Institute of Carbon Neutrality, Peking University, Beijing 10871, China; orcid.org/0000-0002-7731-5399

Zichong Chen – School of Engineering and Applied Science, Harvard University, Cambridge, Massachusetts 02138, United States

Complete contact information is available at:
<https://pubs.acs.org/10.1021/acs.est.4c09822>

Author Contributions

R.L. and Y.Zhang designed the study. R.L. performed inverse modeling. W.C. performed the initial concentration fields and boundary conditions for the simulation. R.L. and Y.Zhang interpreted the results with assistance from J.X., Y.Zhao, P.Z., M.Z., G.S., and Z.C. Q.H. and T.L. performed the CH4MOD simulation. S.L. and W.Y. performed the IBIS simulation and provided flux measurements at Jiansanjiang. J.X. compiled and analyzed rice growth stage data. R.L. and Y.Zhang wrote the paper with contributions from all authors.

Notes

The authors declare no competing financial interest.

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